

Using Actor-Partner Interdependence Models (APIM) to explore indistinguishable and distinguishable dyads with Multilevel (MLM) & Structural Equation Modeling (SEM) approaches.

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Workshop Structure Overview

01. Foundations

Understand dyadic dependencies, assumptions, key APIM parameters, and structural distinguishability.

02. Indistinguishable

Model exchangeable partners using Multilevel Modeling (lme4) and SEM equality constraints.

03. Distinguishable

Evaluate systemic roles (e.g. gender, hierarchy) using moderator, two-intercept, and multi-group SEM variations.

04. Advanced Topics

Address moderated crossover, diary designs, sample size guidelines, and missing data using FIML.

What Is Dyadic Data?

Dyadic data arise when observations are gathered in pairs (dyads) that demonstrate systematic interdependence. Common examples include:

- ✓ **Romantic Couples:** Partners in heterosexual or same-sex relationships (often distinguishable).
- ✓ **Workplace Dyads:** Supervisor-subordinate, mentor-mentee configurations.
- ✓ **Parent-Child Pairs:** Mothers & daughters, fathers & sons.
- ✓ **Peer Groups:** Roommates, sports teammates, co-workers.

⚠ *Non-independence violates classic OLS regression assumptions, leading to biased standard errors and highly inflated Type I error rates.*

📊 Simulated Workshop Dataset

*We utilize a double-earner dataset styled after **Hahn, Binnewies, & Dormann (2014, JVB)**:*

- ✓ **Sample Size:** $N = 100$ couples (200 unique individuals)
- ✓ **Focal Predictor (X):** Work-Nonwork Connection (WNC)
- ✓ **Secondary Covariate (Z):** Psychological Recovery
- ✓ **Target Outcome (Y):** Relationship Satisfaction
- ✓ **Nesting Indices:** Gender, children indicators

Workshop Exercises

An independent simulated dataset for practising crossover-of-work-outcomes techniques on a coherent research question.

Research Scenario

How does a partner's affect, self-determination, and job crafting crossover to shape an actor's work outcomes?

- ✓ **Predictors:** actor & partner affect, SDT, job crafting
- ✓ **Outcomes:** engagement, performance, creativity
- ✓ **Moderators:** live together, years together, time spent this morning

Dataset Specification

N = 250 dual-earner dyads (500 individuals)

- Gender: male / female
- ICC(1) predictors ≈ 0.30

Files: exercise_data.Rdata ddl2 (long), ddw2 (wide)

Focal crossovers:

partner job crafting x live_together → engagement

partner SDT x years_together → performance

partner affect x time spent → creativity

Exercises & Answer Template

A guided, prose-only workflow for adapting scripts 02-08 to the exercise dataset - emphasis on interpretation, not copy-paste.

exercises/exercises.md

9 sections mirroring scripts 02-08 + integrative Section 9.

- ✓ **Sections 1-3:** *scripts 02-04 (APIM, multilevel, distinguishable)*
- ✓ **Sections 4-6:** *scripts 05-07 (distinguishable SEM, two-intercept)*
- ✓ **Section 9:** *integrative reflection across all models on the exercise dataset*

exercises/answers_template.R

Zero executable code — R functions are described in prose, never called. Students adapt scripts 02-08 via a substitution table (ddl2→ddl, affect→wnc, etc.).

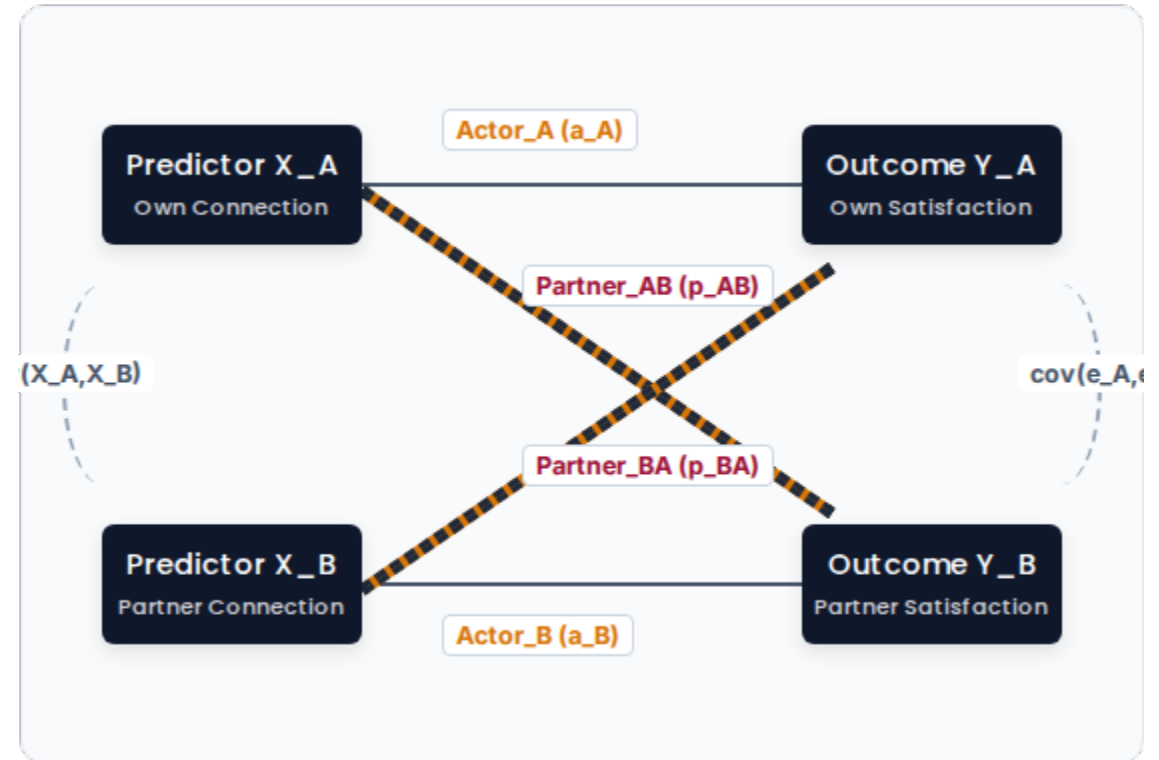
How to use the template:

1. *Run simulate_exercise_data.R to load data*
2. *Read exercises.md section by section*
3. *Open answers_template.R alongside each script*
4. *Apply the substitution table to map data to scripts*
5. *Write a 1-sentence interpretation as a comment*

The APIM Framework

The **Actor-Partner Interdependence Model** (Kenny, Kashy, & Cook, 2006) resolves dyadic dependencies by estimating individual pathways simultaneously:

- ✓ **Actor Effects (a):** The influence of your own predictor status on your own target outcome (e.g. Own connection → Own satisfaction).
- ✓ **Partner Effects (p):** The interpersonal crossover of your partner's predictor on your outcome (e.g. Partner's connection → Own satisfaction).
- ✓ **Simultaneous Control:** Modeling actor and partner paths together extracts true cross-partner influence while isolating individual biases.



Indistinguishable vs. Distinguishable

Indistinguishable Dyads

Members are exchangeable. No baseline or stable structural variable differentiates them.

- ✓ **Arbitrary Labels:** Partner labels (e.g. A vs B, Left vs Right) are completely interchangeable.
- ✓ **Equality Constraints:** Actor and partner slopes must be constrained to be mathematically equal across partners.
- ✓ **Typical Examples:** Same-sex roommates, co-workers of equal organizational rank, sibling pairs.

Distinguishable Dyads

Members can be classified using a systematic, theoretical variable that is consistent across the sample.

- ✓ **Systematic Labels:** Partners are labeled using a meaningful criterion (e.g. Wife/Husband, Boss/Employee).
- ✓
- ✓ **Typical Examples:** Heterosexual couples, medical caregiver-patient dyads, manager-employee pairs.

 **Important:** Indistinguishability is an empirical hypothesis. Always test it formally (by comparing a constrained model with an unconstrained version) before locking in equal-parameter constraints!

Modeling Options Taxonomy

Choose the optimal modeling combination based on your data layout, distinguishability, and nested structures:

Modeling Strategy	Analytical Framework	Data Layout	Core Application & Mechanism
Indistinguishable APIM	MLM (e.g., lme4)	Long Format	Symmetric pooling of slopes across members; simple intercept ICC check.
Moderated MLM APIM	MLM (e.g., lme4)	Long Format	Incorporates member interactions directly into continuous pathways.
Two-Intercept MLM APIM	MLM (e.g., lme4)	Long Format	Suppresses overall intercept to map distinct group means simultaneously.
Standard SEM APIM	SEM (e.g., lavaan)	Wide Format	Strict parameter constraints, goodness-of-fit indices, FIML for missing data.
Multilevel SEM (MSEM)	MSEM (e.g., lavaan)	Long (Nested)	Simultaneous latent variables and complex nesting within nested arrays.



Framework Choice: MLM models excel at complex nesting patterns, daily diaries, and unbalanced structures. SEM provides fit indices, simultaneous constraints, and mediation tests.



Specification Tip: Wide-format SEM allows testing of complex, non-linear partner spillover parameters and structural residuals that are difficult to write in traditional MLM notation.

Multilevel Modeling (Long Format)

To handle indistinguishable datasets, format the file in long format (one row per person, two rows per dyad). Random effects adjust for nested interdependence.

- ✓ **Empty Intercept-Only Model:** Compute the Intra-class Correlation (ICC) to estimate the proportion of shared variance.
- ✓ **Model Specification:** Estimate both actor and partner paths symmetrically, pooling slope estimators across the dyad members.
- ✓ **Centering Strategy:** Group-mean center independent variables to cleanly separate within-dyad from between-dyad effects.

MLM Formulation

$$\text{satisfaction}_{ij} = (\beta_0 + u_j) + \beta_1(\text{wnc}_{ij}) + \beta_2(\text{partner_wnc}_{ij}) + e_{ij}$$

Where i represents the individual, j represents the dyad, u_j represents the dyad-level random intercept, and e_{ij} is individual residual.

>— Requires the *lme4* package in R for model fitting, and *lmerTest* to compute Satterthwaite degrees of freedom.

Reshaping Dyadic Data

↔ Wide Format (1 Row/Dyad)

Standard for SEM with suffix variable labels mapping to specific roles.

dyad_id	wnc_a	wnc_p	sat_a	sat_p
101	4.2	3.8	5.1	4.9
102	3.1	3.2	4.8	4.7

```
# Reshape from Wide to Long
long_df ← wide_df %>%
  pivot_longer(cols = -dyad_id,
    names_to = c(".value", "role"),
    names_sep = "_")
```

↕ Long Format (2 Rows/Dyad)

Standard for MLM with repeated observations nested within a single key.

dyad_id	role	wnc	sat
101	person_1	4.2	5.1
101	person_2	3.8	4.9
102	person_1	3.1	4.8
102	person_2	3.2	4.7

```
# Reshape from Long to Wide
wide_df ← long_df %>%
  pivot_wider(id_cols = dyad_id,
    names_from = role,
    values_from = c(wnc, sat))
```

R Code: Indistinguishable MLM

We estimate the basic APIM framework step-by-step using lmer.

Ensure your package environment is properly configured:

- ✓ Loads data in **long format** (one row per partner).
- ✓ Calculates the **ICC** to verify within-dyad correlation.
- ✓ Includes covariates: Recovery, Children status, and Dual-Earner category.

i Note: In indistinguishable setups, standard slope estimators are pooled across groups.

```
# Indistinguishable MLM (script 02)
library(lme4); library(lmerTest)
load("dyad_data.RData")

# 1. Null model + ICC
m0 <- lmer(sat ~ 1 + (1 | dyad), data=ddl)
icc(m0) # between / (between + residual)

# 2. Full APIM
m1 <- lmer(sat ~ wnc + partner_wnc +
           recovery + has_children +
           dual_earner + (1 | dyad), data=ddl)
summary(m1)
```

Hands-On: Indistinguishable MLM

Active Code Session

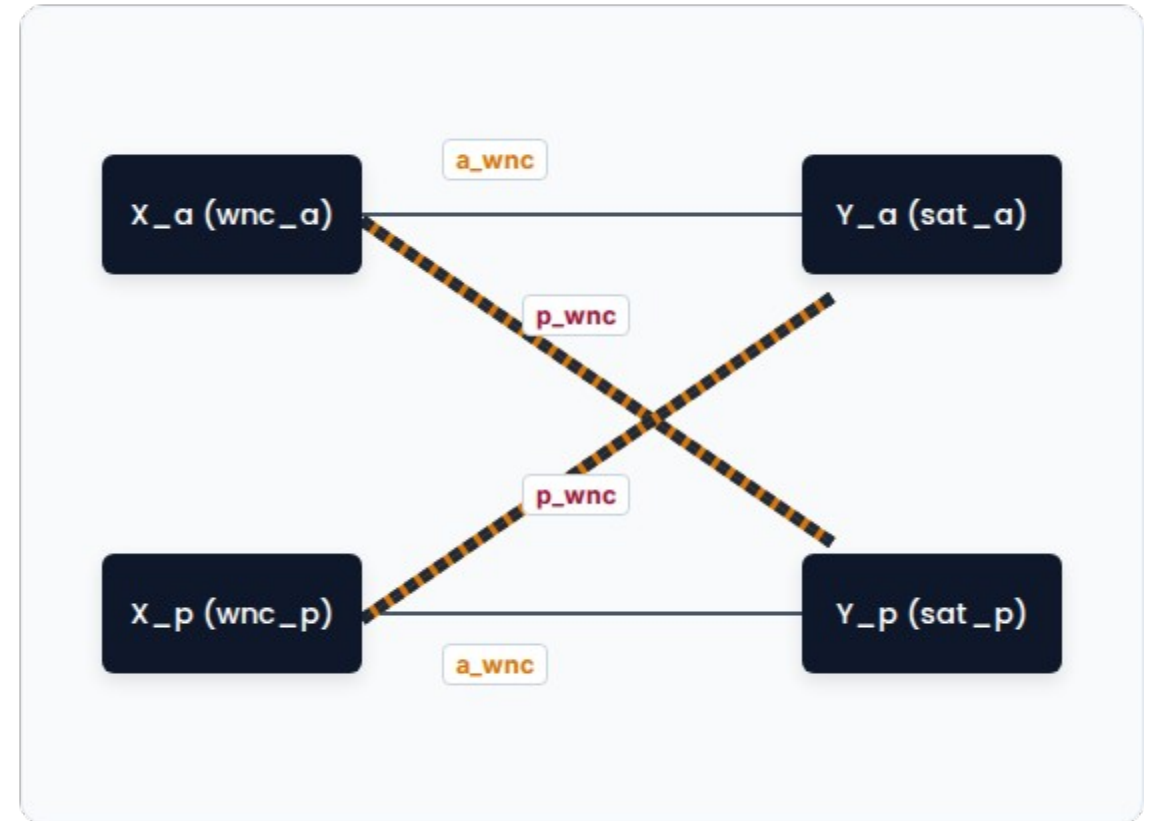
Let's execute your first multilevel APIM scripts in RStudio:

- ✓ **Open:** *scripts/02_indistinguishable_mlm.R*
- ✓ **Run:** *Estimate the null model and compute the ICC:
 $ICC = \sigma^2_{\text{between}} / (\sigma^2_{\text{between}} + \sigma^2_{\text{residual}})$*
- ✓ **Review:** *Run the full APIM and note the actor vs partner effect signs.*

Structural Equation Modeling

We specify explicit path lines inside wide-format data (one row per dyad with distinct suffix labels *_a* / *_p*). Equality constraints are declared to enforce indistinguishability:

- ✓ **Slope Constraints:** Own paths (*b_a*) and partner paths (*b_p*) are labeled identically.
- ✓ **Interceptions/Variances:** Model intercepts, variances, and residuals are restricted to equality.



R Code: Indistinguishable SEM

We write structural code using lavaan syntax. Suffix tags label identical parameters across equations to enforce symmetry:

- ✓ **Label Syntax:** Prefixing coefficients with label names (e.g. $a_{wnc} * wnc_a$) forces equality.
- ✓ **Interceptions & Variances:** Restricting intercepts ($\alpha * 1$) and residuals ($var_res * sat_a$) establishes full indistinguishability.

```
# Indistinguishable SEM wide (script 03)
library(lavaan)
mod_c <- '
  sat_a ~ a_wnc*wnc_a + p_wnc*wnc_p +
          a_rec*rec_a + p_rec*rec_p +
          c_ch*child + c_du*dual_earner
  sat_p ~ a_wnc*wnc_p + p_wnc*wnc_a +
          a_rec*rec_p + p_rec*rec_a +
          c_ch*child + c_du*dual_earner
  sat_a ~ alpha*1
  sat_p ~ alpha*1      # equal intercepts
  sat_a ~~ var_res*sat_a
  sat_p ~~ var_res*sat_p # equal residuals
'
fit <- sem(mod_c, data=ddw)
```

Hands-On: Indistinguishable SEM

Active Coding Session

Let's transition from multilevel modeling to structural equation modeling:

- ✓ **Open:** `scripts/03_indistinguishable_sem.R`
- ✓ **Evaluate:** Review the three alternative analytical paradigms defined in the script:
 - ✓ Model A: Cluster-robust Standard Errors (quick, long format)
 - ✓ Model B: Two-level SEM (within/between decomposition)
 - ✓ Model C: Wide-format APIM (primary, with full equality constraints)

Questions to Explore

Compare the estimated standard errors from Model A (cluster-robust) with Model C (wide-format constraint).



How does structural estimation impact statistical precision in small-to-medium samples?

Core Analytical Insights

Method Convergence

*Balanced data profiles yield identical point estimates in both **Multilevel (long)** and **SEM (wide with equal constraints)** designs.*

LRT Testing

Perform a Chi-Square Difference Test comparing the fully constrained model with an unconstrained version to empirically test for indistinguishability.

Analytical Trade-offs

Why choose SEM over Multilevel for indistinguishable dyads?

- ✓ Supports multi-item latent variables.
- ✓ Supports full mediation models.
- ✓ FIML retains dyads with partial missing values.

Moderator Interaction Modeling

When dyad members are distinguishable (e.g. heterosexual couples), we can model interactions between predictors and the grouping factor (e.g. gender) in long format.

Version A: Simple Moderator (Conflated)

```
satisfaction ~ wnc * gender + recovery * gender + (1 | dyad_id)
```

Version B: Full APIM Moderator (Decomposed)

```
satisfaction ~ wnc * gender + partner_wnc * gender + recovery *  
gender + partner_recovery * gender + (1 | dyad_id)
```

⚠ Warning on Version A

Simple moderator setups (Version A) conflate actor and partner pathways. This can lead to misleading conclusions when evaluating crossover effects.

*Always specify the ****Full APIM Moderator**** (Version B) to properly separate actor from partner pathways when using the moderator approach in long format.*

R Code: Moderator MLM

We write the interaction script for Version B using lmer. This model evaluates if actor and partner effects differ significantly by gender:

- ✓ **Reference Group:** Male is set as the reference group (simple slope = b_{wnc}).
- ✓ **Interaction Term:** Estimates the difference in slopes (female simple slope = $b_{wnc} + b_{wnc:genderfemale}$).

```
# Distinguishable: MLM Moderator (script 04)
library(lme4); library(lmerTest)
load("dyad_data.RData")

# Baseline (no interactions)
m_base <- lmer(sat ~ wnc + partner_wnc +
              rec + p_rec + child + dual +
              (1 | dyad), data=ddl)

# Full APIM with gender interactions
m_mod <- lmer(sat ~ wnc*gender +
              partner_wnc*gender + rec +
              p_rec + child + dual +
              (1 | dyad), data=ddl)

summary(m_mod)
anova(m_base, m_mod) # LRT
```

Hands-On: Moderator APIM

</> Active Coding Session

Let's practice modeling interaction terms in distinguishable dyads:

- ✓ **Open:** `scripts/04_distinguishable_mlm.R`
- ✓ **Analyze:** Fit the full APIM moderator interaction model in R.
- ✓ **Evaluate:** Identify the interaction terms: `wnc:genderfemale` and `partner_wnc:genderfemale`. Calculate the simple slopes for males and females.

💬 Theoretical RQ Example

"Does gender moderate the crossover of work-family connection between spouses, such that the crossover effect of partner connection on relationship satisfaction is stronger for females than for males?"

- ❓ *Are the simple slopes statistically significant for both males and females?*

Distinguishable: SEM Moderator

</> The Approach

Mirror the lme4 gender moderator (script 04) as a two-level lavaan model. Gender interactions enter at the within-dyad level; dyad-level covariates (children, dual earner) at the between level.

💬 lavaan Model

Two-level SEM: gender interactions are placed at the within-dyad level. Numeric coding (1=male, 2=female) makes the simple-slope labels explicit:

$ss_wnc_M := a_w + aw_int * 1$

$ss_wnc_F := a_w + aw_int * 2$

? **Tip:** Models are saturated at the within level, so chi-square difference tests are not available. Use simple-slope labels (:=) to interpret the moderation.

R Code: SEM Moderator

</> Two-Level SEM with Gender Moderation

We construct a two-level lavaan model that mirrors the lmer interaction (script 04):

```
# SEM Moderator (script 05)
library(lavaan)
ddl$gender_num ← as.numeric(ddl$gender)
ddl$wnc_x_g ← ddl$wnc * ddl$gender_num
ddl$pw_x_g ← ddl$partner_wnc * ddl$gender_num

mod ← '
  level: within
    sat ~ a_w*wnc + p_w*partner_wnc +
          ar_w*recovery + pr_w*partner_recovery +
          g_w*gender_num +
          aw_int*wnc_x_g + pw_int*pw_x_g
  level: between
    sat ~ c_ch*has_children + c_du*dual_earner
# Simple slopes
ss_wnc_M := a_w + aw_int * 1
ss_wnc_F := a_w + aw_int * 2
'

fit ← sem(mod, data=ddl, cluster="dyad_id")
summary(fit)
```

The Two-Intercept Model

To avoid setting an arbitrary reference group (like males), we can suppress the global intercept using the 0 + gender syntax.

- ✓ **Direct Parameterization:** Directly estimates intercepts and slopes for both males and females, simplifying group comparisons.
- ✓ **Flexible Equality Constraints:** Easily constrain slopes to be equal across groups while still estimating separate intercepts.

☰ Three Modeling Stages

Compare these nested models using Likelihood Ratio Tests:

- ✓ **Model 1:** Equal intercepts, equal slopes (most constrained model).
- ✓ **Model 2:** Different intercepts, equal slopes (recommended baseline model).
- ✓ **Model 3:** Different intercepts, gender-varying slopes (full interaction model).

R Code: Two-Intercept MLM

The two-intercept model is specified in lme4 by suppressing the global intercept (using 0 + gender) so each group gets its own intercept:

- ✓ **Interpretable Outputs:** Coefficients are presented as separate intercepts (gendermale, genderfemale) rather than standard difference offsets.
- ✓ **Nested Model Comparisons:** Compares nested model equations to identify distinct parameter slopes.

```
# Two-Intercept MLM (script 07)
library(lme4); library(lmerTest)
load("dyad_data.RData")

# Equal slopes, separate intercepts
m2 <- lmer(sat ~ 0 + gender + wnc +
            partner_wnc + rec + p_rec +
            child + dual_earner +
            (1 | dyad), data=ddl)
summary(m2)

# Coefficients: gendermale, genderfemale
# are the two dyad-member intercepts
```

Distinguishable: SEM (lavaan)

In structural equation modeling, distinguishable dyads are modeled using wide-format datasets. We can estimate separate slopes, variances, and covariances for each group without constraints:

- ✓ **Unconstrained Baseline:** Establishes unique pathways (e.g. males: a_m , females: a_f).
- ✓ **Nested Constraints:** Constrain slopes to be equal (e.g. $a == a$) to test for gender differences via a Chi-Square Difference Test.

```
# Distinguishable SEM wide (script 06)
library(lavaan)
# Three nested models:
# 1) Unconstrained - all paths free
# 2) Slopes equal - labels a,p,...
# 3) Fully equal - + equal intercepts
mod_un <- '
  sat_a ~ a_h*wnc_a + p_h*wnc_p + ...
  sat_p ~ a_w*wnc_p + p_w*wnc_a + ...
  sat_a ~ int_a*1
  sat_p ~ int_p*1 # separate intercepts
'

fit_un <- sem(mod_un, data=ddw)
lavTestLRT(fit_un, fit_slopes,
           fit_full) # nested LRTs
```


Hands-On: Distinguishable SEM

Active Coding Session

Let's practice setting constraints in structural equation models:

- ✓ **Open:** `scripts/06_distinguishable_sem_wide.R`
- ✓ **Execute:** Fit the unconstrained, slopes-equal, and fully-equal (slopes + intercepts) models.
- ✓ **Compare:** Run Chi-Square Difference Tests comparing the three models to determine if constraining male and female slopes to be equal significantly worsens model fit.

Core Modeling Decisions

"If the Chi-Square Difference Test is non-significant, adopt the constrained model for parsimony. If significant, retain the unconstrained model to allow for gender differences."



Does model fit change significantly?

Distinguishable: Model Comparison

Compare the three primary structural modeling methods used for distinguishable dyad analyses:

Feature	Moderator Approach (lme4)	Two-Intercept (lme4)	SEM Approach (lavaan)
Data Format	Long Format (One row/person)	Long Format (One row/person)	Wide Format (One row/dyad)
Intercepts	Reference Group + Offset	Two Direct Intercepts	Two Direct Intercepts
Testing Slope Differences	Via Interaction Significance	Via Likelihood Ratio Test	Via Chi-Square Difference
Fit Indices	No (relies on log-likelihood)	No (relies on log-likelihood)	Yes (CFI, RMSEA, AIC/BIC)
Missing Data	Listwise Deletion	Listwise Deletion	FIML (retains partial dyads)
Best Application	Testing external continuous moderators	Direct means comparisons	Complex mediation & latent vars

The Buffer Hypothesis

We replicate the theoretical model of ***Hahn, Binnewies, & Dormann (2014)*** to examine if dyad-level moderators alter standard APIM pathways:

- ✓ **Research Question:** Does the presence of children moderate the partner crossover effect (Partner Connection → Own Satisfaction)?
- ✓ **Theoretical Mechanism:** Partner crossover represents stress transmission. Joint domestic demands (e.g. child-rearing) may alter these crossover pathways.
- ✓ **Findings:** The crossover effect is buffered (weaker) for couples with children, as shared family responsibilities alter cognitive work-life boundaries.

📄 Moderated APIM Structure

The model includes interaction terms between the ***Partner Connection*** predictor and the ***Has Children*** indicator.

- i Both MLM and SEM frameworks can accommodate dyad-level moderators.

Multilevel vs. SEM Formulations

We estimate the interaction using either *lme4* or *lavaan*:

- ✓ **lme4 Interaction:**
 $wnc * has_children + partner_wnc * has_children$
- ✓ **lavaan Interaction:** Requires creating manual product indicators for the interactions in wide format before fitting.

 **Open:** `scripts/08_moderated_apim.R`. Run the script and plot the interaction slopes.

```
# Moderated APIM, MLM (script 08)
library(lme4); library(lmerTest); library(interactions)

m_mod <- lmer(satisfaction ~ wnc + partner_wnc +
              has_children +
              wnc:has_children + partner_wnc:has_children +
              (1 | dyad_id), data = ddl)

interact_plot(m_mod, pred = partner_wnc, modx = has_children)

# *** SEM: lavaan ***

# SEM Approach: lavaan (script 08)
ddw$wnc_a_children <- ddw$wnc_a * ddw$has_children
ddw$wnc_p_children <- ddw$wnc_p * ddw$has_children

mod <- '
  sat_a ~ a*wnc_a + p*wnc_p + a_int*wnc_a_children
  sat_p ~ a*wnc_p + p*wnc_a + p_int*wnc_p_children
  actor_no_child := a + a_int * 0
  actor_has_child := a + a_int * 1
'

fit <- sem(mod, data = ddw)
summary(fit)
```

Advanced: APIM Power Analysis

Dyadic power analysis is complex because power is a function of the ***Intra-class Correlation (ICC)***, effect size, and the number of dyads (not individuals).

- ✓ **Basic APIM:** Requires a minimum of 50 dyads; $N = 100\text{--}120$ dyads is preferred.
- ✓ **Partner Effects:** Require larger samples than actor effects due to shared variance.
- ✓ **Moderation:** Cross-level interactions typically require $N > 150$ dyads.

Sample Requirements Metric

$\alpha = 0.05$, power = 0.80 — required N (dyads):

Effect size (β)	$\beta = .15$	$\beta = .25$
Indistinguishable dyads	159	59
Distinguishable dyads	348	129



Estimated with APIMPowerR:

<https://robert-a-ackerman.shinyapps.io/APIMPowerRdis/>

Three-Level Nesting Structures

Dyads are often nested within larger clusters (e.g. organizations, departments, clinic teams), or measured over time:

- ✓ **3-Level syntax in lme4:**
satisfaction ~ wnc + (1 | team_id) + (1 | dyad_id)
- ✓ **Daily Diary / ESM Designs:** *Days nested within Persons, nested within Dyads.*
 - ✓ *Allows modeling of daily actor/partner lag fluctuations.*
 - ✓ *Time-lagged effects: Does yesterday's connection predict today's relationship satisfaction?*

🔑 Bayesian Estimations

For small samples or complex models with multiple random slopes, Bayesian estimations provide stable parameter estimates:

- ✓ **brms:** *Stan-based package with intuitive, lme4-like syntax.*
- ✓ **blavaan:** *Fits Bayesian SEM models.*

Missing Data in Dyadic Designs

If one partner has missing data, listwise deletion in standard regression removes the entire dyad. This significantly reduces statistical power:

- ✓ **SEM FIML:** Full Information Maximum Likelihood estimates parameters using all available data under MAR assumptions. This is the preferred approach.
- ✓ **Multiple Imputation:** Impute missing values at the dyad level using packages like mice to preserve dyadic dependencies.

✚ Analytical Decision Tree

- ✓ **Step 1: Test Distinguishability:** Compare constrained and unconstrained models via a Chi-Square Difference Test.
- ✓ **Step 2: Choose Your Framework:** Use MLM for longitudinal diary designs or complex random slopes. Use SEM for latent variables, fit indices, or complex mediation.

Questions & Discussion

Thank You!

We have successfully walked through the foundations, estimations, and advanced extensions of the Actor-Partner Interdependence Model (APIM) in R.



R Workspace Datasets & Scripts: Check the local scripts/directory to run exercises.

Primary References

-  Kenny, D. A., Kashy, D. A., & Cook, W. L. (2006). *Dyadic Data Analysis*. Guilford Press.
-  Hahn, V. C., Binnewies, C., & Dormann, C. (2014). *Journal of Vocational Behavior*, 84(2), 137–148.